MIL-P-46320C(MU)
1 APRIL 1963
SUPERSEDING
MIL-P-46320B(MU)
2 JULY 1962

MILITARY SPECIFICATION

POWER SUPPLY: 10516158 (TRANSISTORIZED)

1. SCOPE

1.1 This specification covers one type of transistorized, miniaturized, 16,000 VOlt direct current power supply, capable of being energized by a low-impedance direct current 1.5 volt primary power source.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-290 – Nickel Plating (Electrodeposited)

MILITARY

MIL-F-13926 – Fire Control Materiel: General Specification Governing the Manufacturing and Inspection of

MIL-P-14232 – Parts, Equipment and Tools for Ordnance Materiel, Packaging of

MIL-I-45208 – Inspection Requirements; General Specification for

STANDARDS

MILITARY

MIL-STD-105 – Sampling Procedures and Tables for inspection by Attributes

MIL-STD-202 –Test Methods for Electronic and Electrical Component Parts

DRAWINGS

MUNITIONS COMMAND

C10516158 –Power Supply: 10516158

PACKAGING DATA SHEET

MIL-P-14232/P10516158 – Packaging of Power Supply: 10516158

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The power supply furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

FSC 1240

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3.2 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2.1 External surfaces. The external metallic case of the power supply shall be nickel plated in accordance with Class I of Specification QQ-N-290, except that plating shall have a minimum thickness of 0.0003 inch.

3.3 Construction. The power supply shall be manufactured in accordance with Drawings C10516158 and all drawings and specifications pertaining thereto.

3.3.1 Drawings. The drawings forming a part of this specification are outline drawings intended to show the configuration of the power supply under procurement. Critical dimensions and tolerances are furnished, and no deviation from dimensions or tolerances specified on the applicable outline drawing shall be permitted without prior approval of the contracting officer.

3.4 Output terminal. The circular convolutions surrounding the power supply high voltage output terminal shall mate with the high voltage connector assembly.

3.5 High voltage surfaces. High voltage surfaces shall exhibit a well rounded polished surface free from burrs or sharp edges.

3.6 General specification. The contractor shall be responsible for the following requirements of MIL-F-13926:

(a) Order of precedence

(b) Dimensions and tolerances

(c) Inorganic protective surface finishes

(d) Part identification and marking

(e) Workmanship

3.7 Environmental.

3.7.1 Thermal shock. The power supply shall show no evidence of damage and shall meet the requirements of this specification after exposure to the ambient temperatures specified below within plus or minus 5° Fahrenheit.

(a) The power supply shall be subjected to a gradual increase in ambient temperature until plus 160° Fahrenheit is attained, and maintained at that ambient for two hours. The ambient temperature shall then be lowered to plus 125° Fahrenheit for one hour. The power supply shall then be immediately exposed' to standard ambient temperature (60° to 90° Fahrenheit).

(b) The power supply shall be subjected to a gradual decrease in ambient temperature until minus 65° Fahrenheit is reached and maintained at that ambient for two hours. The ambient temperature shall then be raised to minus 25° Fahrenheit for one hour. The power supply shall then be immediately exposed to standard ambient temperature.

3.7.2 Operating temperatures. The power supply housing shall be thermally stabilized at an ambient temperature of plus 125°, plus 5° minus 0°, Fahrenheit. It shall then be energized and continuously operated' at the above temperature, without a heat sink, for at least eight hours. While still thermally stabilized at the above temperature the power supply shall meet applicable requirements of 3.9. The power supply housing shall be thermally stabilized at an ambient temperature of minus 40°, plus 0° minus 5°, Fahrenheit and then operated for at least one hour. While still thermally stabilized at this latter temperature the power supply shall meet the applicable requirements of 3.9.

3.7.3 Mechanical shock. The power supply shall show no evidence of damage and shall

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meet the following requirements of this specification after exposure to shock impacts of 25 gravitational units (G's) applied six times parallel with and six times perpendicular to the power supply's longitudinal axis at standard ambient temperature.

3.7.4 Mechanical vibration. The power supply at standard ambient temperatures shall not be damaged when subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second (cps), and return to 10 cps, and shall be traversed in approximately one minute, This motion sholl be applied for a period of 10 minutes in each of three mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the power supply shall meet the following requirements of this specification.

3.7.5 Hermetic sealing. When thermally stabilized at an ambient temperature not in excess of 100 degrees Fahrenheit, the power supply, shall exhibit no evidence 'of compound or air leakage, when immersed (for no less than two minutes, and no more than three minutes) in a bath of water maintained at a minimum ambient temperature of plus 180 degrees Fahrenheit.

3.8 Power supply performance; electrical.

3.8.1 Input polarity. The power supply shall meet the requirements of 3.9 when its metallic housing is polarized negative and considered as electrical ground with respec: to its positive input terminal.

3.8.2 Input immunity. The power supply shall not be damaged by application of its rated input operating voltage in reverse polarity.

3.8.3 Input impedance. The power supply shall operate from a direct current (dc) power source whose output impedance is compatible to that of a single 1.5 volt (series C) type BA42 dry cell battery.

3.8.4 Output polarity. The power supply metallic housing shall be considered electrical ground and shall be positive with respect to its negative high voltage output terminal.

3.8.5 Short-circuit protection. The power supply shall not be damaged or adversely affected as a result of a direct short circuit applied for at least one minute during operation between the negative high voltage output terminal and housing.

3.8.6 Intermittent operation. The power supply shall exhibit no evidence of spasmodic or interrmittent operation when meeting the requirements of 3.9.

3.9 Power supply performance; operational.

3.9.1 Input voltage. Unless otherwise specified' herein, the power supply shall meet the applicable operational requirements of this specification when operated from a direct current (dc) low impedance source of 1.52 volts (see 6.5).

3.9.2 Input current. The input current in dc milliamperes (ma) drawn by the power supply shall not exceed the values specified in tables I to III inclusive when delivering the prescribed output voltages at the indicated ambient temperatures.

3.9.3 Output characteristics. The power supply shall demonstrate the following output characteristics at the ambient temperatures specified,

3.9.3.1 Output voltage regulation. The power supply shall deliver a dc output voltage, within the prescribed limits under the four load current conditions specified in Tables I to III inclusive at the indicated ambient temperatures.

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3.9.3.2 Output voltage relative to input voltage. At standard ambient temperature, the power supply shall deliver the minimum value of dc output voltage specified in Table IV upon application of the prescribed value of dc input voltage under the three indicated load current conditions.

TABLE I *Electrical characteristics (+ 125°F.)*

Minimum output current (amperes)	Maximum input current (Ma)	Minimum output voltage excluding dc ripple (KV)
1×10^{-9}	24.0	15.8
1×10^{-7}	26.0	15.5
5×10^{-7}	30.0	15.0
1×10^{-6}	35.0	14.0

TABLE II. *Electrical characteristics (+60° to +90°F)*

Minimum output current (amperes)	Maximum input current (Ma)	Minimum output voltage excluding dc ripple (KV)
1×10^{-9}	18.0	15.8
1×10^{-7}	20.0	15.5
5×10^{-7}	24.0	15.0
1×10^{-6}	30.0	14.0

TABLE III. *Electrical characteristics* (-40°F.)

Minimum output current (amperes)	Maximum input current (Ma)	Minimum output voltage excluding dc ripple (KV)
1×10^{-9}	24.0	15.8
1×10^{-7}	26.0	15.5
5×10^{-7}	30.0	15.0
1×10^{-6}	35.0	14.0

TABLE IV. Output vs input voltage *(+60° to +90°F)*

Maximum dc input Volts	Minimum kilovolt output at dc load currents specified		
	1×10^{-9} amperes	1×10^{-7} amperes	5×10^{-7} amperes
1.00	10.6	10.4	10.1
1.30	13.6	13.4	13.1
1.50	15.6	15.4	15.0
1.52	15.8	15.5	15.1
1.55	15.8	15.6	15.3

3.9.3.3 Maximum output voltage. The maximum output voltage including the peak ripple voltage shall not exceed 16 Kilovolts (KV) under any condition.

3.9.3.4 *Rise time.* The prover supply shall deliver 15KV dc under a load current of 5 X 10-7 amperes in not more than 10 sec. ends, immediately after application of its input voltage at the ambient temperatures specified in 3.7.2.

3.9.3.5 *Ripple.* The peak to peak ripple voltage of the power supply shall not exceed one percent of its output voltage at the ambient temperatures specified in 3.7.2.

3.10 Operability. The power supply shall not be damaged or adversely affected and shall meet the requirements of 3.9 after being subjected at standard ambient temperature to one hour of continuous operation followed by the electrical shock imposed by one hour of intermittent operation. Intermittent operation shall consist of one minute cycles; with each cycle consisting of 30 seconds "on" and 30 seconds "off." The power supply output terminals shall be shorted out during each "off" portion of the cycle.

3.11 Minimum operating life. The power supply, excluding battery, shall meet all requirements of this specification after completion of 1000 hours of continuous operation at standard ambient temperature.

3.12 High voltage breakdown. The power supply shall exhibit no evidence of corona, arcing, or spraying when meeting the requirements of 3.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory

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acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed' necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Except where otherwise specified herein, the contractor's inspection system shall be in accordance with MIL-I-45208.

4.3 Classification of tests. Inspection and testing shall be classified as follows:

(a) Qualification tests
(b) Acceptance tests

4.3.1 Qualification tests. Qualification tests shall consist of tests for all the requirements of this specification. These tests shall be performed in a Government laboratory by Government personnel.

4.3.1.1 Specimens for test. A total of three specimens shall be submitted for qualification approval. The specimens shall be finished products representative of the materials and workmanship used in the manufacture of the items to be furnished under this specification.

4.3.1.2 Qualification approval. Facilities submitting specimens for Qualified Products List (QPL) testing shall be Placed on the QPL if the specimens submitted meet all the conditions of this specification. Failure to meet all of these requirements shall be cause for refusa] to grant qualification approval.

4.3.2 Acceptance tests. Acceptance tests shall consist of an inspection for those characteristics listed in tables V and VI.

4.3.2.1 Final acceptance. Final acceptance of a lot of finished products shall be withheld by-the Government inspector until the sample products, representative of that lot, have satisfactorily met the applicable requirements of this specification.

4.4 Lot formation. When inspection lot sizes and lot formations are applicable, they shall be in accordance with MIL-STD-105 and shall also be defined in the contractor's inspection system.

4.4.1 Defective units and lots. Defective units and lots shall be processed as specified in MIL-STD-105 and shall not be included as a part of the quantity specified in the contract until formal approval is obtained.

4.5 Material. Control of material shall be in accordance with test data requirements of MIL-I-45208.

4.6 Inspection equipment. Supply, calibration, maintenance and use of inspection equipment shall be in accordance with MIL I-45208.

4.6.1 Accuracy requirements. Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except as specified in 4.6.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.2 Inspection equipment design requirements. The inspection equipment used to measure the direct current (dc) input voltage to the power supply shall be capable of measuring that voltage within two percent of the required value. The inspection equipment used to measure the input current drawn by the power supply shall be capable of measuring that current within five percent of the required values.

4.7 Acceptance inspection.

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4.7.1 *Classification of defects and sampling plans.* Examination and testing to determine acceptance or rejection in compliance with the applicable requirements of Section 3 shall be conducted on a 100 percent inspection basis, initially. After conclusive evidence of the process average has been established in conformance with MIL-STD-105 and the AQL's specified herein, the classification of defects and sampling plans specified in tables V and VI may be used in lieu of 100 percent inspection. The AQL value specified shall apply to each individual characteristic on a single defect basis.

4.7.1.1 *Disposition of rejected lots.* Rejected lots shall be screened in conformance to 4.4.1 for all defective characteristics causing rejection of the lot. After the Government inspector has approved the corrective action taken, subsequent production lots may be submitted for acceptance in accordance with 4.7.1 and MIL-STD-105.

TABLE V. Classification *of defects*

Use inspection level II in table III-A with sampling plan table IV-A of MIL-STD-105.

Critical: None defined

Major: AQL 0.65%	Requirement Paragraph	QA. paragraph	Test method
101. Construction, nonconforming	3.3	4.8.1.1	Visual-Tactile-Standard Measuring Equipment (SME)
102. External surfaces, nonconforming	3.2.1	4.8.1.2	SME-Visual
103. Physical failure, post vibration	3.7.4	4.8.1.6	Visual-Tactile
104. Input polarity, incorrect	3.8.1	4.8.1.9	SME
105. Input impedance, incorrect	3.8.3	4.8.1.11	Special Testing Equipment (STE) -SME
106. Output polarity, incorrect	3.8.4	4.8.1.12	SME
107. Intermittent operation	8.8.6	4.8.1.14	STE-SME
108. High voltage breakdown	3.12	4.8.1.16	Visual-Audio
109. Input current excessive	3.9.2	4.8.1.17.1	STE-SME
110. Output voltage regulation error	3.9.3.1	4.8.1.17.2.1	STE
111. Output voltage error	3.9.3.2	4.8.1.17.2.2	STE
112. Maximum output voltage excessive	3.9.3.3	4.8.1.17.2.3	STE

Minor: None defined

Note. The above tests shall be conducted at standard ambient temperature (plus 60° to plus 90° Fahrenheit) in accordance with the Quality Assurance (QA. par.) listed.

TABLE VI. *Classification of defects*

Use inspection level L6 in table III-B with sampling plan table IV-A of MIL-STD-105.

Critical: None defined

Major AQL 4.0%	Requirement paragraph	QA.paragraph	Test method
101. Input current excessive	3.9.2	4.8.1.17.1	Special Testing Equipment (STE). Standard Measuring Equipment (SME).
102. Output voltage regulation error	3.9.3.1	4.8.1 .17.2.1	STE
103. Maximum output voltage excessive	3.9.3.3	4.8.1 .17.2.3	STE
104. Rise time excessive	3.9.3.4	4.8.1 .17.3	STE-SME
106. Ripple excessive	3.9.3.5	4.8.1 .17.4	SME
106. Leakage (sealing)	3.7.5	4.8.1.8	Visual

Minor. None defined

Note. The above teste shall be conducted at ambient temperature of minus 40″ and Plus 186″ Fahrenheit in accordance with the quality assurance paragraph (QA. Par.) listed, except the sealing teat which shall be conducted at standard ambient temperature subsequent to the operating test.

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4.8 Test methods and procedures.

4.8.1 *Qualification tests.*

4.8.1.1 ConStruction. The power supply shall be subjected to a visual tactile and dimensional inspection for conformance to the drawing and requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.8.1.2 *External surfaces.* The external metallic case of the power supply shall be visually inspected for completeness of the specified finish. Thickness measurements of plating will be made by use of standard measuring equipment. Destructive testing, when necessary for QPL analysis, shall be performed using standard measuring equip - ment.

4.8.1.3 *Minimum operating life.* The power supply shall be subjected to a minimum of 1000 hours of continuous operation and then subjected to the following tests and shall conform to the requirements specified in 3.11.

4.8.1.4 *Thermal shock.* This test shall be performed in test chambers that shall maintain the power supply at the required tem-Perature for the time periods indicated. Subsequent to each specified temperature cycle, high and low, the power supply shall be subjected to a visual and tactile examination and shall show no evidence of damage.

4.8.1.5 *Operating temperatures.* This test shall be conducted in accordance with Procedure I of MIL-F-13926, except that the temperature shall not be reduced beyond minus 40° or raised beyond plus 125° Fahrenheit. While stabilized at each temperature extreme the power supply shall be energized and continuously operated for the time period specified. At the completion of this operating period the power supply shall be subjected to the tests specified in 4.8.1.17.1, 4.8.1.17.2.1, 4.8.1.17.2.3, 4.8.1.17.3, 4.8.1.17.4 and shall conform to the requirements specified.

4.8.1.6 *Mechanical shock.* This test shall be conducted in accordance with Method 202 of MIL-STD-202. Subsequent to the shock test the power supply shall be subjected to a visual and tactile examination and shall show no evidence of physical failure. The power supply shall then be subjected to the following tests and shall conform to the requirements specified in 3.7.3.

4.8.1.7 *Vibration.* This test shall be conducted in accordance with Method 201 of MIL-STD-202, except that the motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Performance measurements of the power supply are not required during vibration testing. Subsequent to vibration, the power supply shall be subjected to a visual and tactile examination and shall show no evidence of physical failure. The power supply shall then be subjected to the following tests and shall conform to the requirements specified in 3.7.4.

4.8.1.8 *Hermetic sealing.* The power supply when at a temperature not to exceed 100° Fahrenheit shall be immersed for 2 to 3 minutes in a bath of water, or any other suitable liquid of no greater viscosity, maintained at a temperature of at least 180° Fahrenheit. During this test the power supply shall be considered to have met the specified performance requirement when visual examination provides no evidence of air leakage, resulting in a continuous flow of air bubbles, or compound leakage.

4.8.1.9 *Input polarity.* This test shall be conducted with standard measuring equipment. Input polarity of the power supply shall conform to the requirements specified in 3.8.1.

4.8.1.10 *Input immunity.* This test shall be conducted by applying a minus 1.52 volts

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direct current to the input terminal of the power supply. Subsequent to the application of this voltage the power supply shall show no evidence of damage, and shall meet the requirements specified in 3.8.2.

4.8.1.11 *Input impedance.* Energize the power supply with a 1.5 volt (Series c) type BA42 dry cell battery and apply an output load current of 1 x 10^{-6} to the power supply. The minimum measured output voltage of the power supply shall be 14.0 KV direct current. Voltage measurements shall be measured with standard measuring equipment.

4.8.1.12 *Output polarity.* This test shall be conducted with standard measuring equipment. Output polarity of the power supply shall conform to the requirement specified in 3.8.4.

4.8.1.13 *Short-circuit protection.* When energized, the power supply shall be subjected to a directed short circuit for the time period specified. After removal of the short circuit, the power supply shall show no evidence of damage, and shall meet the requirements specified in 3.8.5.

4.8.1.14 *Intermittent operation.* The power supply shall be observed for spasmodic or intermittent operation during the tests specified in 4.8.1.17. Any evidence of spasmodic or intermittent operation shall be cause for rejection.

4.8.1.15 *High voltage breakdown.* The power supply shall be observed for high voltage breakdown during the tests specified in 4.8.1.17. Any evidence of corona, arcing or spraying shall be cause for rejection.

4.8.1.16 *Operability.* The power supply shall be subjected to the cycling operation specified. Upon completion of the cycling operation, the power supply shall be subjected to the subsequent tests and shall meet the requirements specified in 3.10.

4.8.1.17 *Input voltage.* The following tests shall be conducted with the specified input voltage applied to the input terminal of the power supply. The low impedance power source supplying the input voltage shall be connected through a control device that shall be capable of maintaining a constant input voltage to the power supply. Unless otherwise specified, the input voltage to the power supply shall be 1.52 volts dc (direct current) during all tests. These tests shall be performed in a controlled atmosphere to avoid any adverse affect of temperature and humidity on the test circuitry. A suggested method for minimizing the effects of humidity on the test circuitry is shown on figure 2.

4.8.1.17.1 *Input current.* This test will be conducted concurrent with tests specified in 4.8.1.17.2.1. The measured current drawn by the power supply shall not exceed the milliampere value specified for the temperature indicated, when the input voltage and output load current to the power supply is varied as required. Input current shall be measured by use of standard measuring equipment.

4.8.1.17.2 *Output characteristics.* Output characteristics shall be tested with a special testing device. A suggested method of construction of the device is shown in figures 1 and 2. The device shall contain a means for varying the input voltage as required.

4.8 .1.17.2.1 *Output voltage regulation.* This test shall be conducted when the power supply is thermally stabilized at the specified temperatures. When the value of output load current is as indicated, the measured output voltage of the power supply shall be within the applicable values shown in table I through III.

4.8.1.17.2.2 *Output voltage relating to input voltage.* When the input voltage to the power supply is varied under the three output load conditions indicated in table IV, the measure output voltage of the power

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supply for each condition shall meet the minimum value as specified.

4.8.1.17.2.3 Maximum output voltage. The ripple voltage value obtained in 4.8.1 .17.4 shall be added to each output voltage value obtained in 4.8.1.17.3. Each sum of output and ripple voltage shall not exceed the maximum voltage value specified in 3.9.3.3.

4.8.1 .17.3 *Rise time.* Rise time shall be measured by use of a standard calibrated time measuring instrument and the special testing device specified in 4.8.1.17.2. Rise time of the power supply shall meet the requirements specified in 3.9.3.4.

4.8.1.17.4 *Ripple.* Ripple voltage shall be measured with standard measuring equipment. A suggested method of measurement is shown in figure 1. The measured ripple voltage shall not exceed the tolerance specified in 3.9.3.5.

4.8.2 *Acceptance test.* The power supply samples selected in accordance with 4.7.1 shall be Subjected to the tests specified in 4.8.1.1, 4.8.1.2, 4.8.1.7, 4.8.1.8, 4.8.1.9, 4.8.1.11, 4.8.1.12, 4.8.1.14, 4.8.1.15 and 4.8.1.17.1 through 4.8.1.17.4 inclusive. The power supply shall meet the requirements specified.

4.8.3 *Preservation, packaging and packing.* Quality assurance provisions for preservation, packaging and packing shall be in accordance with MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10516158. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The power supply covered by this specification is intended to supply the electrical power required to operate infrared image tubes, or other items requiring a miniaturized transistorized device capable of delivering microampere of dc in the kilovolt range.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of an applicable level of protection.

(c) Applicable stock number.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the U. S. Army Munitions Command, Frankford Arsenal, Philadelphia 37, Pa., Attn: SMUFA-5500, and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions.

6.4.1 *Standard measuring equipment (SME).* Standard measuring equipment (SME) as defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

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6.4.2 *Special testing equipment (STE).* Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.4.3 Tactile. Skillfull manipulation or Sensing.

6.5 Caution. The application of an excessive power supply input voltage, gradually or instantaneously may result in permanent damage to power supply components.

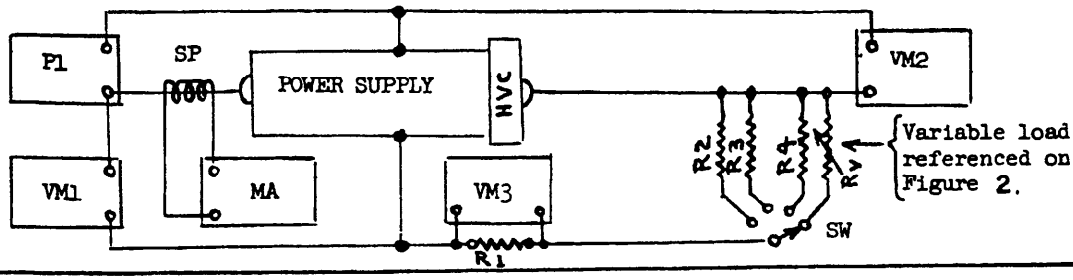
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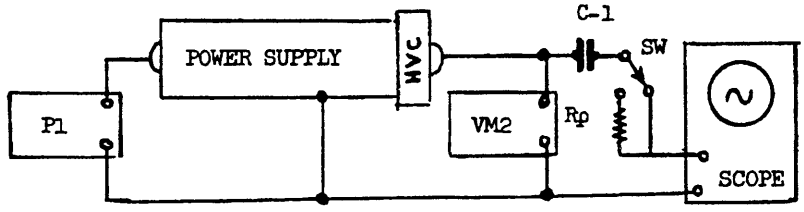
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METHOD OF MAKING INPUT-OUTPUT MEASUREMENTS



METHOD OF MAKING RIPPLE VOLTAGE MEASUREMENTS



SCOPE 1.0 megohm internal impedance
Rp 10.0 megohm probe
C-1 0.001 microfarad dc blocking capacitor

PROCEDURE:

1. Read peak to peak voltage on scope using 10 megohm probe, record as V_1
2. Read peak to peak voltage without probe, record as V_2
3. Calculate:

$$\text{Ripple Voltage} = V_1 \left[\left(\frac{V_2 - V_1}{V_1 - 10V_2} \right) + 1 \right]$$

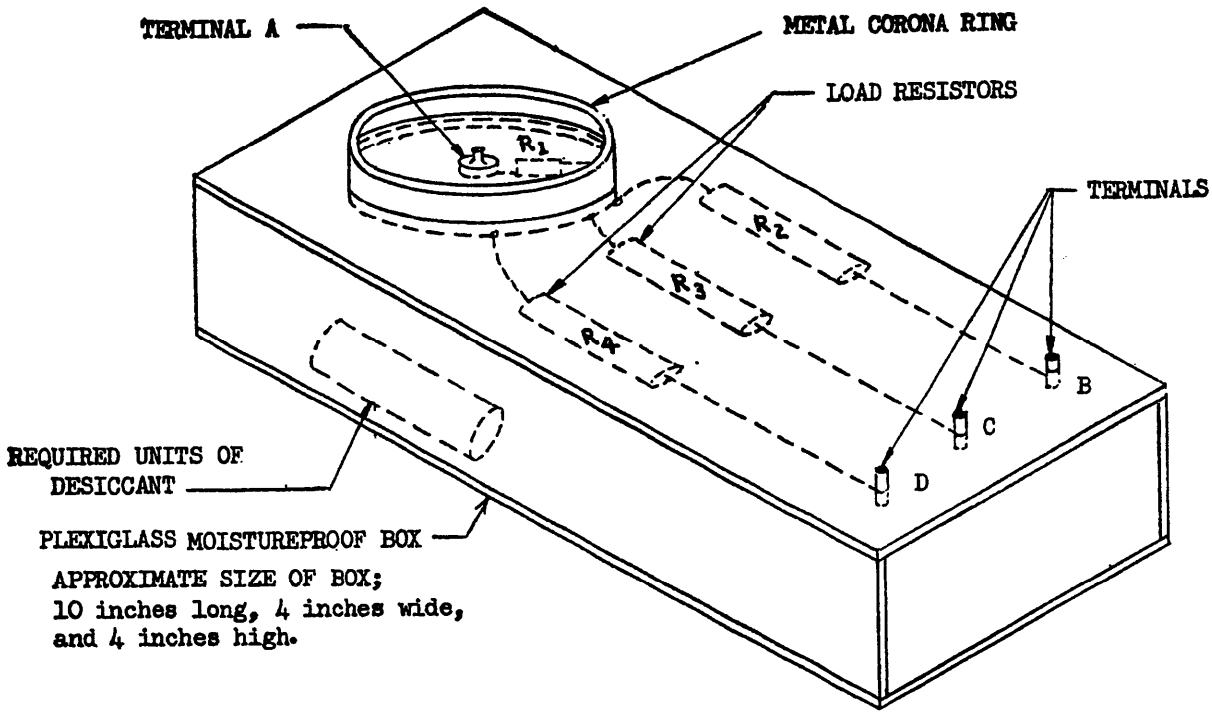
TEST EQUIPMENT

P1	Regulated (Tubeless) Low Impedance Power Supply, Electronic Research Associates, Inc., Model 110DMC or equal	
VM2	Electrostatic voltmeter 0-30KV, Sensitive Research Mod. ESH or equal	
VM1	DC Voltmeter, 20,000 ohm-per-volt	
VM3	Vacuum Tube Voltmeter, Hewlett-Packard Model 410B or equal	
MA	DC Milliameter, Hewlett-Packard Model 428A or equal	
SP	Sensing probe, supplied with above dc milliameter	
HVC	High voltage connector, American Research & Manufacturing Co., Model HS or equal	
SW	Low loss switch, should withstand 30KV	
R1	10 megohms, 2% Type HBR Resistor	Resistance Products Co. or equal
R2	160,000 megohms, 5% Type HBR Resistor	
R3	16,000 megohms, 5% Type HBR Resistor	
R4	32,000 megohms, 5% Type HBR Resistor	

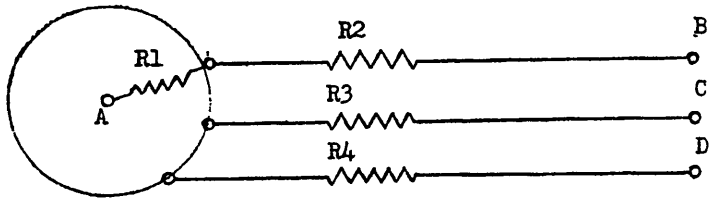
NOTES:
1. Extreme care should be taken to make sure that the High Voltage Connector (HVC) is free of foreign matter.
2. High voltage test leads should withstand 30KV
3. Extreme care should be taken to guard against the adverse effects of humidity on test wiring and components
4. Load resistor Rv is used when testing under a variable load
5. The voltage drop across R1 is utilized to determine load current.

FIGURE 1. *Suggested test methods.*

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NOTE: Value of R1, R2, R3, and R4 same as shown on Figure 1. Each individual resistance may consist of more than one resistor to minimize leakage currents.



NOTE: A continuously variable load may be obtained when a type 6914 electron tube and voltage controlled light source, to irradiate the tube photocathode, are both mounted within a lightproof box in lieu of the above load resistors.

FIGURE 2. *Suggested method of constructing a load box to minimize the effects of humidity on output voltage measurements.*

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